

Subgame Perfection and Win-Lose Games

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Subgame Perfect Equilibrium

- Last time we solved sequential games using **backward induction**.
- Today we refine Nash equilibrium using this idea: the **subgame perfect equilibrium**.
- This lets us identify which threats are credible and which are empty.
- We'll also look at **win-lose games**, where backward induction yields clean results.

Associated Reading

Game Theory, Sections 3.4 and 3.5

Recap: The Entry Deterrence Game

Entrant

├ Stay Out $\rightarrow (0, 4)$

└ Enter

└ Incumbent

├ Fight $\rightarrow (-1, 1)$

└ Accommodate $\rightarrow (2, 2)$

Payoffs: (Entrant, Incumbent)

By backward induction: the Incumbent would Accommodate ($2 > 1$), so the Entrant Enters ($2 > 0$). Outcome: (Enter, Accommodate) with payoffs (2, 2).

Why the Incumbent's Threat Fails

The Incumbent might *prefer* the Entrant to Stay Out (they get 4 instead of 2).

Could the Incumbent **threaten** to Fight if the Entrant enters?

The threat would require fighting, which gives the Incumbent payoff 1. But a rational Incumbent prefers 2 (Accommodate).

The threat is **not credible**. A rational Entrant knows the Incumbent won't carry it out.

This is one of the main insights of backward induction: it reveals which threats and promises are credible.

Subgame Perfect Equilibrium

What Is a Subgame?

A **subgame** is any part of the game tree that:

- Begins at a single decision node.
- Contains all branches following from that node.
- Is a complete game in itself.

In the entry game: the whole game is a subgame, and so is the Incumbent's decision node after entry.

Subgame Perfect Equilibrium

A **subgame perfect equilibrium** (SPE) is a strategy profile that constitutes a Nash equilibrium **in every subgame**.

This rules out Nash equilibria that rely on **non-credible threats or promises** off the equilibrium path.

In perfect information games: the backward induction outcome is always the unique SPE.

A Nash Equilibrium that isn't Subgame Perfect

In the entry game, consider the strategy profile:

- Entrant: **Stay Out**
- Incumbent: **Fight** (if entered)

Is this a Nash equilibrium?

Yes. Given the Incumbent will Fight, the Entrant's best response is to Stay Out ($0 > -1$). And the Incumbent's strategy doesn't matter since the node is not reached.

But it's **not subgame perfect**. At the Incumbent's subgame, Fight is not a best response (Accommodate gives $2 > 1$).

The Importance of Subgame Perfection

Subgame perfection refines Nash equilibrium by requiring rationality **everywhere** in the game tree, not just on the equilibrium path.

It eliminates equilibria held in place by empty threats.

This makes it the dominant solution concept for perfect information sequential games.

More Examples

The Centipede Game

Two players alternate between *continuing* (C) or *stopping* (S) the game. Each stage, the pot grows, but the player who stops takes more.

At each node, stopping gives the current player a larger share than continuing, *if* the other player is about to stop.

Backward Induction in Centipede

Apply backward induction from the end:

- At the last node: Player 1 stops (gets more).
- At the second-to-last node: Player 2 knows P1 will stop, so P2 stops.
- At each earlier node: same logic applies.
- **Result:** Player 1 stops immediately, at the very first move.

The backward induction prediction is that no cooperation ever happens, even though both players would do much better if they cooperated longer.

The Paradox of Backward Induction

The Centipede Game shows a tension:

- **Backward induction** predicts immediate stopping.
- But this requires Player 1 to reason: “If I reach move 3, Player 2 is rational... but how did we reach move 3?”

In experiments, people rarely stop immediately. Cooperation often unfolds for many rounds.

Whether this shows backward induction is *wrong*, or merely that people are *not* perfectly rational, is a substantive debate.

The Chain Store Game

Setup

A chain store operates in **10 towns**. In each town, a local firm decides whether to **Enter** or **Stay Out**.

If a local firm enters, the chain store can **Fight** (price war) or **Accommodate** (share the market).

Payoffs in each town are the same as the entry deterrence game:

Outcome	Entrant	Chain Store
Stay Out	0	4
Enter + Fight	-1	1
Enter + Accommodate	2	2

The chain store's **total** payoff is the sum across all 10 towns.

The Chain Store's Temptation

The chain store would prefer every local firm to Stay Out (payoff 4 per town, total 40).

Could it **threaten** to Fight any entrant, to deter the others?

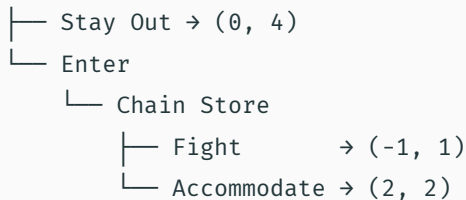
This seems plausible. Fighting in Town 1 might convince Towns 2–10 that entry is costly.

But backward induction says this threat is never credible. Let's see why.

Backward Induction: Town 10

Start at the **last town**. There are no future towns to deter.

Entrant 10



This is just the entry deterrence game. The chain store prefers Accommodate ($2 > 1$). **Entrant 10 enters.**

Now consider Town 9. The chain store knows that **whatever happens in Town 9**, Entrant 10 will enter and the chain store will accommodate in Town 10.

So fighting in Town 9 yields no future benefit — it cannot change Town 10's outcome.

The chain store again prefers to Accommodate in Town 9 ($2 > 1$). **Entrant 9 enters.**

The same logic applies to every town, working backwards:

- Town 10: Fighting yields no future deterrence. Chain store accommodates.
- Town 9: Fighting cannot affect Town 10's outcome. Chain store accommodates.
- Town 8: Same reasoning. Chain store accommodates.
- $\vdots$
- Town 1: Same reasoning. Chain store accommodates.

Every local firm enters. **Every** entry is accommodated. The chain store earns 2 per town — total 20, not 40.

Why a Known Endpoint Matters

The unraveling argument depends on one critical assumption: **the number of towns is common knowledge**.

Backward induction starts at Town 10 *because everyone knows Town 10 is last*. Without a known endpoint, there is no base case for the induction.

If the chain store faces an **infinite** sequence of potential entrants — or even just an **unknown** number — the argument fails. There is no “last town” to start from, so the backward induction never gets going.

In that setting, fighting early entrants *can* be part of an equilibrium, because the threat of future fighting is never undermined by a known terminal node.

Why the Threat Unravels

The chain store's threat, "I will Fight any entrant", is **not credible in any town**, because:

- Credibility in Town k requires that fighting there deters future entrants.
- But deterrence only works if the chain store will fight in Towns $k + 1, \dots, 10$.
- Backward induction shows it won't fight in those towns either.

Each link in the chain of deterrence breaks, so the whole chain breaks.

The Subgame Perfect Equilibrium

The unique subgame perfect equilibrium is:

- **Every entrant:** Enter.
- **Chain store:** Accommodate in every town.

The chain store's threat to Fight is a Nash equilibrium strategy (if entrants stay out, the fight node is never reached), but it is **not subgame perfect**, just like in the single-town entry game.

This result is striking because the intuition for fighting seems strong:
“If I fight in the early towns, later entrants will be deterred. And the cost of fighting is small compared to the benefit of 10 towns of monopoly profits.”

Yet backward induction rules this out entirely. The game-theoretic prediction is stark.

In experiments, chain stores **do** fight early entrants, and deterrence **does** seem to work, at least for a while.

Whether this means players are irrational, or that backward induction is too demanding a standard, or that this isn't the right model, is an open question.

The Chain Store Game was introduced by Reinhard Selten (1978), who called the conflict between intuition and backward induction the **Chain Store Paradox**.

Selten himself found the backward induction result hard to accept as a description of real behavior, despite having proven it.

One resolution: if there is even a **small probability** that the chain store is the type that always fights (perhaps for non-monetary reasons), the equilibrium changes dramatically. Even a small probability can be enough to get to the deterrence equilibrium. The first working out of this was the **reputation model** built by David Kreps and Robert Wilson in 1982. (If you do this stuff enough, every interesting game theory question has either a contribution by Selten or by Kreps; this one has both.)

Win-Lose Games

Consider two-player perfect-information games where the only outcomes are “Player 1 wins” or “Player 2 wins.” We assign payoffs $(1, 0)$ and $(0, 1)$ respectively.

These **win-lose games** have a remarkably clean structure when analyzed by backward induction.

An Example: The Number Game

Two players take turns choosing a number from $\{1, 2, \dots, 10\}$. Player 1 goes first. The player whose choice brings the running total to 100 or more wins.

Player 1 has a winning strategy. By backward induction, the “losing positions” (where the player whose turn it is cannot win) are: 89, 78, 67, 56, 45, 34, 23, 12, 1.

Player 1 starts with 1. Then, whatever Player 2 picks (call it n), Player 1 picks $11 - n$. This brings the total to the next losing position for Player 2 each time.

Theorem (Zermelo, 1913): In every finite two-player win-lose game with perfect information, one of the two players has a **winning strategy**.

The proof uses backward induction. At the root, either some branch leads to a $(1, 0)$ outcome (Player 1 can win), or all branches lead to $(0, 1)$ outcomes (Player 2 wins regardless).

What About Draws?

Games like chess and checkers allow three outcomes: Player 1 wins, Player 2 wins, or Draw.

An extension of Zermelo's theorem says exactly one of three cases holds:

- Player 1 has a strategy guaranteeing a win.
- Player 2 has a strategy guaranteeing a win.
- Both players have strategies guaranteeing at least a draw, so with optimal play the outcome is a draw.

Checkers falls into the third category (proven by computer in 2007). For chess, we still do not know which category applies.

Three Types of Solution

This page is taken from the useful Wikipedia page on solved games.

Ultra-weak solution

Prove whether the first player will win, lose or draw from the initial position, given perfect play on both sides.

Weak solution

Provide one algorithm for each of the two players, such that the player using it can achieve at least the optimal outcome, regardless of the opponent's moves, from the start of the game.

Strong solution

Provide an algorithm that finds optimal plays for both players from all legal positions.

In symmetric games that allow *passing*, it is often simple to provide an ultra-weak proof that player two can't win.

If there is a winning strategy for Player 1, they can play it.

If there is no winning strategy for Player 1, they can pass, and now Player 2 can't win.

The first player can win.

This was first shown back in 1988.

The first full working out of the game tree, and of optimal moves in all situations, was last year, in 2025. These strong solutions are really hard to build.

A side effect of this was showing who (if anyone) can force a win from an arbitrary board position.

Worked Example

A Regulation Game

A company (Player 1) decides whether to **Pollute** (cheaper production) or go **Clean**. If the company pollutes, a regulator (Player 2) can **Fine** the company or **Ignore** the violation.

Company



Payoffs: (Company, Regulator)

Part 1: Backward Induction

Step 1: At the Regulator's node, Fine gives 1 and Ignore gives 0. The Regulator chooses **Fine**.

Step 2: The Company anticipates this. Clean gives 2; Pollute gives 0 (since the Regulator will Fine). The Company chooses **Clean**.

Backward induction outcome: Clean, with payoffs (2, 3).

Part 2: A Non-Credible Threat?

Suppose the Regulator announces: “We will Ignore any pollution.”

If the Company believed this, they would Pollute (payoff 4 vs. 2). The outcome would be (Pollute, Ignore) with payoffs (4, 0).

Is (Pollute, Ignore) a Nash equilibrium?

Check: Given Ignore, the Company prefers Pollute ($4 > 2$). Given Pollute, does the Regulator prefer Ignore? No, Fine gives $1 > 0$. So this is **not** a Nash equilibrium. The Regulator's promise to ignore violations is not even self-consistent.

Part 3: What About (Clean; Fine)?

Consider: Company plays Clean, Regulator plays Fine.

Given the Regulator will Fine, Company prefers Clean ($2 > 0$). Given the Company plays Clean, the Regulator's strategy doesn't affect the outcome (the Fine node is never reached). So the Regulator has no reason to deviate.

This **is** a Nash equilibrium, and it's also subgame perfect. In the subgame starting at the Regulator's node, Fine is the best response ($1 > 0$).

Part 4: Could (Clean; Ignore) Be a Nash Equilibrium?

Given the Regulator plays Ignore, Company prefers Pollute ($4 > 2$). So Company would deviate.

This is **not** a Nash equilibrium. The Regulator's lax policy only works as a strategy if the Company never actually tests it, but the Company *would* test it.

This example shows why backward induction and subgame perfection matter: they force us to check what happens at decision nodes that are “off the equilibrium path.”

- We move to games where players may **randomise** by choosing strategies with probability.
- This requires cardinal payoffs (not just ordinal rankings).
- And it opens up a new kind of Nash equilibrium: **mixed strategy equilibrium**.